Business Fraud Detection, Prevention and Investigations with Conversational AI

Dr. Hrishikesh Desai

Assistant Professor Accounting / Director of the Master of Accountancy with Data Analytics Program
Neil Griffin College of Business, Arkansas State University, USA
Email: hdesai@astate.edu

ABSTRACT:

As financial crimes evolve, traditional fraud detection systems struggle to keep pace. This chapter explores how the latest conversational AI technologies like ChatGPT or Claude can revolutionize fraud management. Leveraging natural language processing, these AI assistants can analyze data intelligently, identify red flags, provide regulatory context, and guide investigations. The seven case studies in this chapter illustrate conversational AI's applications across anti-money laundering (AML) processes - transaction monitoring, network analysis, alert suppression, sanction screening, know-your-customer (KYC), audits, and regulatory reporting. The AI assistants can detect complex patterns, visualize criminal networks, rationalize alerts, automate filings, and enable auditors to validate AI model fairness transparently. However, responsible implementation of conversational AI in fraud management requires addressing data quality, privacy, AI bias, and model explainability challenges. Techniques like data masking, SHAP/LIME, counterfactual explanations, and fairness metrics are proposed as potential solutions. Conversational AI empowers financial institutions with cognitive capabilities to outmaneuver fraudsters continuously. Yet ethical AI principles of accountability, transparency and fairness must be upheld for successful adoption.

KEYWORDS:

1. Anti-Money Laundering
2. Bias Mitigation
3. ChatGPT
4. Claude
5. Conversational AI
6. Data Privacy
7. Explainable AI
8. Fraud Detection

1. Introduction

As financial crimes and fraudulent activities continue to evolve in sophistication, traditional rule-based systems are proving insufficient in keeping pace with the ever-changing tactics employed by bad actors (Association of Certified Fraud Examiners, 2023). Fortunately, advanced artificial intelligence (AI) and natural language processing (NLP) technologies have significantly improved cognitive computing, enabling more effective detection, prevention, and investigation of financial fraud. NLP is a branch of AI that helps computers understand, interpret, and generate human language. In this chapter, I discuss how conversational AI can be used for fraud detection, prevention, and investigations using seven case studies. I also explore how financial institutions and regulatory bodies can leverage the power of conversational AI solutions to bolster their anti-money laundering (AML) and fraud prevention efforts. AML includes measures designed to prevent the concealment of illegally obtained money.

Recent technological advancements in artificial intelligence have led to the development of large language models and conversational AI systems that hold significant potential in fraud management. Notable examples include ChatGPT and Codex (developed by OpenAI), Claude (developed by Anthropic), Copilot (developed by Microsoft and GitHub), and Gemini (developed by Google). These AI systems represent a substantial leap forward in natural language processing capabilities, enabling them to engage in human-like dialogues, understand complex queries, and provide detailed, context-aware responses. While it may be premature to label them as 'innovative' in the long-term sense, their current applications in fraud detection and prevention certainly represent novel approaches in the field.

This chapter is structured as follows: First, I provide an overview of how conversational AI can be applied to various aspects of AML compliance, including transaction monitoring, network analysis, alert suppression, sanction screening, and know-your-customer (KYC)

processes. To illustrate the practical applications of conversational AI in fraud detection, prevention, and investigation, I present seven illustrative scenarios. These scenarios are not formal case studies but rather hypothetical situations designed to demonstrate how conversational AI can be applied in various aspects of fraud management.

The seven illustrative scenarios covered in this chapter are:

1. *Transaction Monitoring:* Analyzing suspicious wire transfers using AI-assisted pattern recognition.
2. *Network Analysis:* Mapping complex relationships between entities in potential money laundering schemes.
3. *Alert Suppression:* Reducing false positives in transaction monitoring alerts.
4. *Sanction Screening and KYC:* Enhancing customer due diligence processes with AI assistance.
5. *System Audits and Healthchecks:* Using AI to improve the auditing of fraud prevention systems.
6. *Regulatory Compliance:* Automating Suspicious Activity Report (SAR) filings.
7. *Managing AI Challenges:* Addressing data quality, privacy, bias, and explainability issues in AI-powered fraud prevention systems.

Each scenario demonstrates how conversational AI can be leveraged to improve efficiency, accuracy, and effectiveness in different aspects of fraud management. While the scenarios presented are hypothetical, they are grounded in real-world challenges faced by financial institutions in fraud detection and prevention. The applications of conversational AI demonstrated in these scenarios reflect emerging trends in the field, but it's important to note that actual implementations may vary based on specific institutional needs, regulatory environments, and technological capabilities.

Following these scenarios, I discuss the challenges associated with implementing conversational AI in fraud prevention systems, including data quality and privacy concerns, bias mitigation, and the need for explainable AI. I conclude by addressing these challenges and offering insights into the future of AI-powered fraud management. Throughout the chapter, I emphasize the importance of responsible AI implementation and adherence to regulatory requirements.

2. Conversational AI for AML Compliance

AML compliance is a critical aspect of financial crime prevention, involving a range of activities such as transaction monitoring, network analysis, alert suppression, sanction screening, and know-your-customer (KYC) processes. KYC is a process by which businesses verify the identity of their clients and assess potential risks of illegal intentions. First, it's important to understand this jargon related to financial frauds.

Transaction monitoring refers to continuous tracking and analysis of financial transactions to identify any suspicious or unusual activity that could potentially be linked to money laundering or other financial crimes.

Network analysis involves mapping out the relationships and connections between various indivdiuals or entities involved in financial transactions. This helps uncover hidden links, identify potential criminal networks, and trace the flow of funds across different parties, which can be crucial in detecting money laundering schemes.

Alert suppression is the process of intelligently filtering out any false alerts of fraud cases, reducing the noise and allowing investigators to focus on the truly suspicious cases. Due to the vast volume of transactions and the complexity of 'fraud detection' rules, transaction monitoring systems often generate a large number of alerts, many of which may be false positives (legitimate transactions flagged as suspicious).

Sanction Screening is the process of checking individuals, entities, and transactions against government-issued sanctions lists or watchlists to ensure that financial institutions do not inadvertently facilitate transactions with sanctioned parties.. These lists contain names of individuals, organizations, and countries that are subject to economic or trade sanctions due to their involvement in activities such as terrorism, drug trafficking, or human rights violations.

Know Your Customer (KYC) refers to the due diligence processes that banks and financial institutions undertake to verify the identity, background, and risk profile of their customers. This includes gathering and verifying customer information, understanding the nature of their business, and assessing the potential risks associated with the customer relationship. Robust KYC procedures are crucial for detecting and preventing financial crimes, as they help institutions avoid dealing with high-risk or suspicious individuals or entities.

Traditionally, these tasks have been carried out using rule-based systems and manual interventions, which can be time-consuming, error-prone, and inefficient in dealing with the ever-increasing volume and complexity of financial data.

Conversational AI, powered by natural language processing capabilities, has revolutionized AML compliance by enabling intelligent and intuitive interactions with financial data and regulatory requirements. In this chapter, I present seven case studies covering various use cases on how ChatGPT and other conversational AI assistants can be used in the context of fraud detection, prevention and investigations. These case studies have been created by me based on some of the potential use cases of conversational AI in fraud suggested to me by H3M Analytics, Inc (2023) and Flagright (2024).[1]

[1] H3M Analytics, Inc. leverages AI in its KROTON compliance product suite, offering solutions for transaction monitoring, network analysis, alert suppression, sanction screening, KYC processes, and audits. Flagright's AI Forensics also represents a groundbreaking approach to anti-money laundering (AML) investigations, harnessing the power of conversational AI to enhance efficiency, accuracy, and speed in fraud detection processes. AI Forensics integrates seamlessly into Flagright's centralized platform, providing a user-friendly interface that supports natural language queries. This innovative feature also enables investigators to input questions or commands in plain language, significantly reducing the learning curve and allowing for more intuitive data exploration. As per Flagright, the core of AI Forensics lies in its ability to process and analyze vast amounts of

2.1 Conversational AI & Transaction Monitoring

By engaging with ChatGPT or similar conversational AI assistants, compliance professionals can query and analyze vast amounts of data, identify potential red flags, and receive contextual insights and recommendations in real-time. For instance, ChatGPT can be trained on AML regulations, typologies (here: common patterns, methods, or techniques used by criminals to launder money or finance illegal activities), and historical case studies, allowing it to provide relevant guidance and flag suspicious activities or patterns.

Illustrative Scenario 1: Analyzing Transactions

This hypothetical scenario demonstrates how conversational AI can enhance the process of analyzing suspicious wire transfers, particularly in the context of potential corporate embezzlement or money laundering.

Suppose a financial institution has detected a series of wire transfers from a corporate account to multiple personal accounts in different countries, totaling $500,000 over a three-month period. This could potentially be a case of money laundering or illegal fund diversions. In this scenario, ChatGPT and other conversational AI assistants could be trained on:

1. AML Regulations:

    a) Bank Secrecy Act (BSA) requirements for reporting suspicious transactions. BSA is a U.S. law requiring financial institutions to assist government agencies in detecting and preventing money laundering.

    b) FATF (Financial Action Task Force) recommendations on money laundering typologies. FATF is an intergovernmental organization that develops policies to combat money laundering and terrorism financing.

data from diverse sources, including transaction records, customer profiles, and external databases (Flagright, 2024).

2. Typologies:
    a) Structuring/smurfing (breaking up large transactions into smaller ones to avoid reporting)
    b) Tax evasion through offshore accounts
    c) Trade-based money laundering
    d) Layering
3. Case studies:
    1. Historical cases of corporate fraud/embezzlement
    2. Money laundering schemes involving shell companies

With this training, the AI system would be able to quickly analyze complex transaction patterns and compare them against specific, relevant typologies and regulations. Also, when an investigator inputs the details of the wire transfers, ChatGPT and other conversational AI assistants would be able to:

1. Identify that the activity potentially matches the typology of fund diversion/embezzlement based on transfers from a corporate account to personal accounts across borders. For example, the AI system could flag these activities as instances of "layering" in the money laundering process, where funds are moved through multiple accounts to obscure their origin.
2. Cite relevant AML regulations like the BSA requirement to file a Suspicious Activity Report (SAR) for such transactions. SARs are required to be filed with the Financial Crimes Enforcement Network (FinCEN), which is a bureau of the U.S. Department of the Treasury that collects and analyzes information about financial transactions to combat financial crimes, when money laundering or fraud involving over $5,000 is

suspected. SAR is a document that financial institutions must file with the FinCEN following a suspected incident of money laundering or fraud.[2]

3. Pull up similar historical cases, like the XYZ company embezzlement scandal, to provide context. For example, the AI system could reference a similar historical case, such as the "ABC Corp Embezzlement Case of 2019," where a Chief Financial Officer (CFO) diverted company funds to personal offshore accounts.
4. Recommend further investigation steps like analyzing corporation's financials, checking personal account holders against watchlists, tracing origins of funds etc.
5. Provide risk scoring on the likelihood of the activity being suspicious based on patterns from the training data. For example, the AI system could provide a risk score of 8/10 for the likelihood of these activities being suspicious.

In this manner, conversational AI assistants can act as an intelligent guide, connecting different dots, providing legal/regulatory context, and aiding the human investigator in comprehensively analyzing the potentially suspicious activity. As a result, the investigation process can be significantly accelerated.

2.2 Conversational AI & Network Analysis

Illustrative Scenario 2: Network Mapping

This scenario demonstrates how conversational AI can assist in mapping and analyzing complex relationships between entities in potential money laundering schemes.

Suppose investigators are analyzing a set of suspicious transactions involving the transfer of large sums between Company A, Company B, and several individual accounts across multiple jurisdictions. On the surface, the transactions seem related to legitimate business dealings. However, conversational AI assistants like ChatGPT or Claude, after being fed this transaction data and trained on typologies, could start finding deeper connections:

[2] Failure to file a SAR can expose the institution to enforcement actions for lack of compliance with the BSA.

1. They could cross-reference the company names and individual account holders against sanctions lists, adverse media reports, and proprietary databases to flag if any entities have been previously linked to financial crimes.
2. Through natural language processing of documentation like contracts, emails, and public records, they could map out the intricate corporate ownerships and relationships between Company A, Company B, and the individual account holders.
3. By analyzing transaction patterns, frequencies, geographic regions involved, they could identify potential discrepancies. For example, a mismatch between the company's stated line of business and the transaction volumes/destinations.
4. They could then visualize and highlight potential networks and fund flows that seem suspicious based on their training. E.g., a complex web of shell companies transferring funds circularly with no apparent economic purpose.
5. For transactions linked to specific high-risk jurisdictions, they could provide contextual information about prevalent terrorist financing methodologies used in those regions based on past case studies.

These conversational AI assistants could then present this comprehensive network mapping, highlighting potential choke points, risk scores, and specific typologies they have detected. This excellent synthesis of disparate data points into key risks and networks of interest can allow human investigators to then focus their efforts on unraveling the intricacies of the financial crime efficiently. The natural language interface allows fraud investigators two-way dialogue to explore further angles as needed.

2.3 Conversational AI & Alert suppression

As discussed earlier, false fraud alerts consume valuable time and resources, hindering the efficiency of fraud prevention efforts. Conversational AI can play a pivotal role in alert suppression by intelligently analyzing the context and risk factors associated with each alert.

By engaging in a dialogue with ChatGPT and other conversational AI assistants, investigators can provide additional context, clarify ambiguities, and receive tailored recommendations on whether an alert should be escalated or suppressed. Moreover, conversational AI assistants can aid in the investigation process by providing fraud investigators with relevant information, synthesizing data from multiple sources, and suggesting potential lines of inquiry based on the specifics of each case. This collaborative approach empowers investigators to make more informed decisions and to better prioritize high-risk cases.

Illustrative Scenario 3: *Identifying False Positives*

This scenario illustrates how conversational AI can help reduce false positives in transaction monitoring alerts, improving efficiency in fraud detection.

Suppose an automated transaction monitoring system at a bank flagged several cash deposits made into a personal account over a period of 6 months, totaling $50,000. This triggered an alert for potential structuring/smurfing activity used in money laundering. The investigator could initiate a dialogue with the conversational AI like ChatGPT or Claude and provide additional context via a prompt such as:

"The account holder is a small business owner dealing in cash sales. The deposits seem to align with the seasonal revenue cycles based on the type of business. However, the deposits were made in amounts just under $10,000 to avoid CTR filings."

CTR (Currency Transaction Report) is a report that financial institutions must file for each deposit, withdrawal, exchange of currency, or other payment or transfer that involves a transaction in currency of more than $10,000. The conversational AI tools, based on their training on AML typologies, regulations like the Bank Secrecy Act CTR requirements, and understanding of business revenue patterns, could analyze the context. They may determine that:

i. While the amounts appear structured, it fits the narrative of a cash-intensive legitimate business.

ii. As long as CTRs are filed for cash transactions over $10,000, no explicit regulation is being violated.

iii. Without stronger adverse information, this may be a false positive not worth pursuing further.

The conversational AI assistants could then recommend suppressing or re-scoring the alert as lower risk based on the evidence provided.

2.4 Conversational AI & KYC

As discussed earlier, it is important that financial institutions do not inadvertently facilitate transactions involving sanctioned entities or individuals with dubious backgrounds. Conversational AI can streamline these processes by intelligently parsing and cross-referencing customer information, transaction details, and sanctions lists. ChatGPT and other conversational AI tools can assist in identifying potential matches, flagging discrepancies, and providing recommendations for further action based on the risk profile of each customer or transaction. Further, they can aid in the Customer Due Diligence (CDD) process, which is a subset of KYC, by engaging in natural language dialogues with customers, gathering relevant information, and assessing risk factors in a more intuitive and user-friendly manner.

Illustrative Scenario 4: *Sanction screening and KYC/CDD*

This scenario shows how conversational AI can streamline and improve the accuracy of sanction screening and Know Your Customer (KYC) processes.

Suppose a bank is onboarding a new corporate client 'Desai Trading, Inc.' and needs to perform thorough due diligence checks. The bank representative can initiate a dialogue with ChatGPT or Claude, providing details like the company name, registered address, ownership information etc. ChatGPT can then:

1. Parse and segment the input data into searchable fields like name, address, and other identifiers.

2. Cross-reference this data against updated sanctions lists from Office of Foreign Assets Control (OFAC), United Nations (UN), European Union (EU), Her Majesty's Treasury (HMT) and other relevant bodies. The OFAC is a financial intelligence and enforcement agency of the U.S. Treasury Department that administers and enforces economic and trade sanctions. The UN is an international organization founded in 1945 to maintain international peace and security. The EU is a political and economic union of 27 member states that are located primarily in Europe. The HMT is the United Kingdom's economics and finance ministry.

3. Check for fuzzy matches (approximate string matching), aliases, former names/addresses using natural language processing.
4. Flag any potential hits or close matches that require further investigation.

This allows comprehensive sanction screening in an intuitive conversational manner. After clearing sanctions, the bank can probe further via ChatGPT or Claude:

"Please walk me through gathering the required KYC information for Desai Trading, Inc. per our policies."

ChatGPT can engage in a dialogue:

1. Request details like business registration, financial statements, ownership structure.

2. Based on the business sector, geography, ask tailored questions to assess risk factors such as "Do you deal in any high-risk goods like arms/minerals?"

3. Clearly explain regulatory requirements and why certain information is needed.

4. Allow uploading/sharing of documents and validating their authenticity leveraging Optical Character Recognition (OCR), blockchain verification etc. OCR is the electronic or mechanical conversion of images of typed, handwritten or printed text into machine-encoded text.

Throughout this back-and-forth, conversational AI tools can provide guidance, explain regulations, flag potential risk indicators for further scrutiny based on their training. The conversational flow and natural language interactions make the KYC process smoother, minimizing back-and-forth, and building rapport with the client while still meeting compliance needs.

Now, in a related scenario about Customer Due Diligence (CDD), suppose the transaction monitoring system flagged outgoing wire transfers from Desai Trading, Inc. in high-risk jurisdictions. The investigator engages ChatGPT or Claude:

"The transfers seem to be recurring payments by an import/export company to suppliers in Panama and UAE for goods received. However, the transaction descriptions and amounts don't match up across the different wire transfers."

The conversational AI assistants could:

- Summarize relevant Financial Action Task Force (FATF) guidance and typologies related to trade-based money laundering using over/under-invoicing of goods.
- Locate any adverse media on the companies/jurisdictions from databases.
- Check if the counterparty entities were present on sanctions lists.
- Analyze the fund flow patterns for any red flags.
- Suggest further steps – review of shipping records, AP/AR documents, & onsite audits.

Thus, this collaborative dialogue allows the AI tools to connect disparate dots, provide targeted information based on the specific case context, and empower the fraud investigator to either quickly resolve or build a comprehensive money laundering investigation successfully. The

natural language interface also makes the AI's knowledge easily accessible and applicable, maximizing human-AI synergy.

2.5 Conversational AI & Auditing

By engaging with ChatGPT and other conversational AI tools, auditors can query the fraud prevention system's knowledge base, understand the rationale behind specific decisions or recommendations, and assess the robustness of the underlying models and data sources. This level of transparency is crucial in building trust and ensuring regulatory compliance. They can also assist in conducting healthchecks by analyzing system logs, identifying potential anomalies or performance issues, and providing recommendations for system optimization and maintenance.

Illustrative Scenario 5: System Audits and Healthchecks

This scenario demonstrates how conversational AI can enhance the auditing and maintenance of fraud prevention systems.

Suppose a regulatory auditor is reviewing the transaction monitoring system at a bank to ensure it is effectively identifying potential money laundering risks and meeting compliance requirements. Conversational AI assistants like ChatGPT or Claude could be directly integrated with a fraud prevention system via APIs.[3] This allows the conversational AI tools to ingest and parse relevant data like: system logs/telemetry data, details of the AI/ML models employed, rules/typologies configured, specific risk scenarios used for transaction monitoring, alert rationalization, knowledge bases used for identifying entities, transactions, etc.

The auditor can then initiate a dialogue with the conversational AI assistants:

[3] If direct system integration is not feasible, the conversational AI could be pre-trained on extensive documentation about the fraud prevention system. This could include: technical documentation of models, rules, data sources; audit/testing reports explaining system logic; sample datasets for training the AI; and compliance documentation mapping the system to regulations. By ingesting and training on this system-specific documentation, the AI can then use natural language processing to provide auditors with relevant insights when queried. Moreover, the conversational AI's responses can be further reinforced by integrating it with real-time monitoring data pipelines, absorbing the latest system log outputs.

"Please explain the logic and data sources behind how this system generates alerts for suspicions of trade-based money laundering."

ChatGPT can respond by walking through:

1. The specific typologies, rules and risk indicators it is trained to detect trade-based laundering cases (e.g. over/under-invoicing, misrepresented goods/services, high-risk jurisdictions etc.)
2. The models and algorithms used to analyze transaction data and identify anomalies
3. The data sources and watchlists integrated (like export/import databases, sanctions lists, adverse media)
4. Illustrate with sample cases from its training data on how it connected the dots to raise alerts

The auditor can further probe by asking it to justify its scoring rationale for real alerts raised during a certain period. Conversational AI tools can provide insights into why certain transactions were deemed high-risk based on the evidence. This transparency builds trust that the system's underlying AI models are operating as intended and aligned with regulatory expectations.

Also, the auditor can leverage ChatGPT's or Claude's natural language capabilities for continuous healthchecks by asking:

"Have you identified any operational issues, degradations or anomalies in the system over the past 6 months?"

These tools can proactively analyze system logs, check for metrics like increasing false positive rates, or uncover component failures which may require attention. It can highlight these findings conversationally. They can also provide recommendations based on reviewing latest cloud infrastructure telemetry data:

"The data volume from the APAC region has increased 35%. You may want to scale up the processing capacity for optimum performance."

The auditor gets this intelligent advisory from the AI in plain language, instead of having to manually inspect and correlate different system data sources. By serving as an interactive interface, the conversational AI assistants allow auditors and compliance teams to effectively validate and continuously monitor the systems in an intuitive manner, while providing clear AI-driven insights when needed.

2.6 Conversational AI & Regulatory Compliance

Conversational AI assistants can be used for automating Suspicious Activity Report (SAR) filings to improve accuracy, completeness, consistency, quality, and alsoachieve significant time savings (H3M Analytics, Inc., 2023).

Illustrative Scenario 6: Automating SAR Filing Processes

This scenario illustrates how conversational AI can streamline the process of filing Suspicious Activity Reports (SARs), addressing implementation challenges, success metrics, and data privacy issues, improving efficiency and consistency in regulatory reporting.

Suppose XYZ Bank, a large financial institution, is required to file Suspicious Activity Reports (SARs) for transactions that exhibit signs of potential money laundering or fraudulent activities. The process of filing SARs involves gathering relevant information, investigating the suspicious activity, and composing a detailed narrative for submission. This manual process is time-consuming and burdensome for the bank's BSA (Bank Secrecy Act) compliance officers, who are responsible for filing SARs within strict regulatory timeframes. Currently, it takes an average of 4 hours to compile and file a single SAR, with a backlog of potential cases growing by 12% monthly.

The bank can implement a conversational AI system, which integrates with the bank's transaction monitoring system, customer database, and case management software and is

trained on historical SAR data, regulatory guidelines, and money laundering typologies. Implementation of such a system might involve a 3-month process: 1 month for system integration, 1 month for staff training, and 1 month for parallel testing.

In addition to detecting suspicious activities and flagging them, the conversational AI assistants can go much further in helping BSA officers:

1. *Automated Information Gathering:* The conversational AI assistants can be asked to automatically collect relevant information about the suspicious transaction from various internal systems, such as customer profiles, account histories, and transaction details. They can also be asked to cross-reference the gathered information with external databases, such as sanctions lists and adverse media, to enrich the data for SAR filing.
2. *Intelligent Narrative Generation:* Using the collected information and their trained knowledge of money laundering typologies, the conversational AI assistants can generate a comprehensive SAR narrative. The narrative would include a detailed description of the suspicious activity, relevant transaction details, customer information, and any additional context or red flags identified. The AI systems can ensure that the narrative is clear and meets regulatory requirements for SAR filing.
3. *BSA Officer Review and Approval:* The generated SAR narrative can then be automatically presented to the BSA compliance officer for review through a user-friendly interface. The BSA officer can review the narrative, make any necessary edits or additions, and provide final approval for filing. The officer can also ask more questions, request additional information, or provide feedback to further refine the narrative.
4. *Automated SAR Filing:* Once the SAR narrative is approved by the BSA officer, the conversational AI systems can be used to automatically populate the required fields in

the SAR filing template. This ensures that all necessary information is included and formatted correctly, minimizing errors and inconsistencies.

XYZ Bank can then measure the success of the AI implementation using the following metrics:

1. *Time Efficiency:* SAR compilation and filing time reduced from 4 hours to 40 minutes on average (83.33% reduction).
2. *Backlog Reduction:* Case backlog decreased by 60% within three months of implementation.
3. *Accuracy:* 96% of AI-generated SARs require minimal to no edits by human reviewers.
4. *Regulatory Compliance:* 100% of SARs filed within required timeframes, up from 86% pre-implementation.
5. *Cost Savings:* Estimated $450,000 annual savings in labor costs and reduced regulatory fines.

As more SARs are prepared and filed using the conversational AI assistants, these systems will continuously learn and adapt based on feedback from BSA officers and regulatory updates. It will also ensure standardized SAR narratives being generated across all filings and minimize risks of non-compliance.

XYZ Bank also addresses data privacy concerns with the help of the AI system through the following measures:

1. *Data Minimization:* The AI only accesses and processes data necessary for SAR filing, adhering to the principle of data minimization.
2. *Encryption:* All data transmissions between systems and to FinCEN are encrypted using Advanced Encryption Standard (AES)-256 encryption, which is a virtually impenetrable symmetric encryption algorithm.
3. *Access Controls:* Multi-factor authentication and role-based access controls limit system access to authorized personnel only.

4. *Data Retention:* SAR-related data is retained only for the legally required period, after which it's securely deleted.
5. *Audit Trails:* Comprehensive logs are maintained of all AI actions and human interactions with the system.
6. *Privacy Impact Assessment:* The bank conducts regular privacy impact assessments to ensure compliance with regulations like GDPR, CCPA, and GLBA.
7. *De-identification:* Where possible, personal data is de-identified or pseudonymized before processing by the AI.

The degree of success of XYZ Bank's AI system would depend on the following factors:

1. *Initial Accuracy:* What is the accuracy of the AI system in the first month? For example, if during the first month, the AI's accuracy was only 65%, more human oversight would be required.
2. *Complex Cases:* Does the AI system struggle with highly complex, unusual cases? If yes, then significant human intervention would still be required.
3. *Regulatory Acceptance:* Is the bank able to demonstrate the AI's reliability to regulators? There might be an initial period of parallel human and AI processing.
4. *Staff Adaptation:* Is there resistance from some to the new system? If yes, then it might necessitate comprehensive change management and training programs to make the AI system implementation successful.

To sum it up, implementing conversational AI systems for SAR automation can significantly improve efficiency and accuracy, but requires careful integration with existing systems and processes. However, success of such systems should be measured across multiple dimensions, including time savings, accuracy, compliance improvements, and cost reductions. Moreover, addressing data privacy is crucial and involves technical measures, policy adjustments, and ongoing assessments. Finally, human oversight may still remain essential, especially for

complex cases and final approvals. Tolga Kurt, Managing Partner at H3M Analytics, Inc., quotes: "*ChatGPT and LLM models in general will significantly enhance the learning curve for new analysts learning to fight financial crime, and minimize their time spend for the related complex investigations.*" (Kurt, 2024).

3. Challenges of using Conversational AI in Fraud Prevention Systems

While the potential benefits of leveraging conversational AI for fraud detection, prevention, and investigation are undeniable, there are several challenges and considerations that must be addressed:

1. **Data Quality and Privacy:** Conversational AI systems rely heavily on the quality and completeness of the data they are trained on. Ensuring data privacy and adhering to relevant regulations regarding data usage and storage is crucial.
2. **Biases:** As with any AI system, there is a risk of inherent biases or opaque decision-making processes (Parikh et al., 2019). Efforts must be made to ensure transparency in the AI models employed.
3. **Explainability:** Explainability in conversational AI refers to the capability of an AI system to provide human-understandable explanations and reasons behind its predictions, decisions or recommendations (Das & Rad, 2020; Dwivedi et al., 2023). In financial fraud detection, explainability is especially important for compliance and auditing, building user trust (if the AI system is a black box), and identifying potential biases in the AI models when they are trained on historical data.
4. **Continuous Learning and Adaptation:** Financial crimes and fraud tactics are constantly evolving (Association of Certified Fraud Examiners, 2023), necessitating continuous learning and adaptation of the conversational AI systems to stay ahead of emerging threats.

Illustrative Scenario 7: *Managing Challenges in AI-Powered Fraud Prevention Systems*

This scenario explores how organizations can address common challenges associated with implementing AI in fraud prevention systems.

A bank has implemented a conversational AI system like ChatGPT or Claude to assist its fraud investigators in analyzing suspicious transaction alerts. The AI assistants are trained on historical transaction data, customer profiles, and past fraud cases. However, there are some concerns with data quality and privacy:

i. The historical transaction data used for training contains a significant number of missing fields and inconsistent formats, affecting the AI's ability to learn accurate patterns.

ii. Some of the customer profile data includes sensitive personally identifiable information (PII) like social security numbers and credit scores, which raises data privacy concerns and potential violations of regulations like the Gramm-Leach-Bliley Act or GLBA, Fair Credit Reporting Act or FCRA, California Consumer Privacy Act or CCPA (if California residents' data is involved), New York SHIELD Act (if New York residents' data is involved), or General Data Protection Regulation or GDPR (if European Union citizens' data is involved) if not handled properly. PII can be any data that could potentially identify a specific individual. The GLBA is a U.S. law that requires financial institutions to explain their information-sharing practices to their customers and to safeguard sensitive data. The FCRA is a U.S. federal law that regulates the collection of consumers' credit information and access to their credit reports. The CCPA is a state statute intended to enhance privacy rights and consumer protection for residents of California. Finally, the GDPR is a regulation in EU law on data protection and privacy in the European Union and the European Economic Area.

Also, the conversational AI system shows some inherent bias and explainability issues:

i. The conversational AI assistants flag a higher proportion of transactions from a certain ethnic group as potentially fraudulent. However, due to the black-box nature of the deep learning models used, fraud investigators are unable to understand why these transactions are being singled out, raising concerns about potential bias.

ii. Investigators are hesitant to trust the AI's recommendations without clear explanations of the reasoning behind each alert, as they need to justify their actions to auditors and customers.

Moreover, there is recent news that has called into question the AI's learning and adaptation abilities:

i. Fraudsters have started using a new money laundering tactic involving cryptocurrencies and online gaming platforms. However, the AI system is not able to effectively detect these new patterns as it was primarily trained on traditional money laundering typologies.

ii. Updating the AI with new training data and fraud scenarios will require significant time and resources from the bank's data science team, creating delays in adapting to emerging threats.

Thus, some of the more experienced fraud investigators may be resistant to using the conversational AI assistants, preferring to stick to their traditional manual investigation methods. The fraud investigators may argue that data quality issues and potential biases leading to a high number of false positives are resulting in a waste of their time on fruitless alerts. They may also have to often second-guess or disregard the system's recommendations due to the lack of explainability. They may also criticize the conversational AI's inability to quickly adapt to new fraud tactics, which leads to missed detection of real money laundering cases, exposing the bank to financial and reputational risks.

In this scenario, to manage all of the above challenges, the bank would need to take some quick steps:

1. Invest in data governance frameworks:
    a) Establish a comprehensive data governance policy that defines data quality standards, metadata management, and tracking of data lineage.
    b) Implement data quality checks and validation rules at the point of data entry and during ETL (Extract, Transform, Load) processes to ensure data completeness, consistency, and accuracy. ETL is a data integration process that combines data from multiple sources into a single, consistent data store.
    c) Use data profiling tools to identify and rectify data quality issues in the historical datasets used for training the conversational AI assistants.
    d) Employ data masking (replacing sensitive data with fictitious but realistic data that maintains the same format and statistical properties as the original data) and tokenization techniques (replacing sensitive data with a non-sensitive surrogate value called a token, and storing the original sensitive data securely in a separate database while mapping it to the corresponding token) to protect sensitive PII data while still allowing the AI to learn patterns from it (Tachepun & Thammaboosadee, 2020)
    e) Regularly audit and monitor data usage to ensure compliance with privacy regulations like GDPR, GLBA, CCPA etc.

For example, in this scenario, since the bank's customer data includes sensitive PII like social security numbers (SSN). Instead of using the real SSNs in the training data, the bank could simply apply one of the following data masking techniques (Shukla et al., 2022):

i. *Shuffling:* Randomly shuffling the digits within each SSN so that the original number is not discernible but the overall format is maintained.

ii. *Substitution:* Replacing the real SSNs with randomly generated numbers that follow the same 9-digit format.

iii. *Encryption:* Encrypting the SSNs using a strong cryptographic algorithm and using the encrypted values for training.

The masked data would then be used to train the conversational AI system. The AI can still learn patterns and correlations from the masked data without being exposed to the actual sensitive values. Alternatively, tokenization could be applied to sensitive data fields like SSNs, credit card numbers, or account numbers.

i. When a new customer record is ingested for training the AI, the SSN is sent to a tokenization system.

ii. The tokenization system would generate a random, unique token (e.g., a string of alphanumeric characters) for that SSN and store the token-to-SSN mapping in a secure database.

iii. The original SSN in the customer record is replaced with the generated token.

iv. The tokenized customer record is then used to train the conversational AI assistant.

When the conversational AI assistant needs to process a transaction or query related to a specific customer, it would use the token instead of the actual SSN. The tokenization system would be queried to retrieve the original SSN if needed for downstream processing, with access controls ensuring only authorized systems or users can de-tokenize the data.

The main benefit of these techniques is that the original sensitive data is never directly exposed to the conversational AI systems, reducing the risk of data breaches or misuse. They can still learn patterns and correlations from the masked or tokenized data, as the format and statistical properties are preserved. It's also easier to comply with data privacy regulations as

the sensitive data is protected. If, in a worst case scenario, the conversational AI system is compromised, the attacker would only have access to the masked or tokenized data, not the actual sensitive information.

2. Implement Explainable AI techniques:
    a) Utilize techniques like SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-Agnostic Explanations) to provide interpretable explanations for the AI's predictions on each suspicious transaction (Alabi, et al., 2023; Neves et al., 2021). SHAP is a game theoretic approach to explain the output of any machine learning model, while LIME explains the predictions of any classifier in an interpretable and faithful manner by learning an interpretable model locally around the prediction.
    b) Integrate the conversational AI with a case-based reasoning (CBR) system that can draw analogies and provide explanations based on similar historical fraud cases (Hillig & Müller, 2021).
    c) Employ counterfactual explanations to show fraud investigators what changes in the input data would have resulted in a different output from the conversational AI assistant (Guidotti, 2022).
    d) Use visual analytics and dashboards to illustrate the key factors and their weightage influencing each conversational AI decision.
    e) Continuously monitor and test the AI models for fairness and bias, using techniques like disparate impact analysis and equality of opportunity metrics (Barocas & Selbst, 2016; Bell et al, 2024; MacCarthy, 2018).

For example, SHAP, based on the concept of Shapley values from cooperative game theory, would assign an importance value for each feature (e.g., transaction amount, location, time) by considering all possible combinations of features and how much each feature contributes to the

prediction in each combination. In the context of fraud detection, the conversational AI assistant would make a prediction on a transaction, classifying it as either fraudulent or legitimate. These Shapley values are then used to explain the prediction by showing the positive or negative contribution of each feature towards the model's output (Molnar, 2022). For example, SHAP might show that the high transaction amount and the unusual location had a strong positive contribution towards classifying a transaction as fraudulent, while the customer's long history with the bank had a negative contribution. These explanations can be visualized using force plots, bar plots, or decision plots, making it easier for investigators to understand the factors driving the AI's predictions (Ariza-Garzón et al., 2020).

On the other hand, LIME tries to understand the AI model's decision-making process by focusing on one specific transaction and its surrounding context (Molnar, 2022). Thus, LIME would take an original transaction and create a new dataset by slightly modifying its features (amount, location, time, the customer's history with the bank, etc.). For example, it might create variations of the transaction with slightly different amounts, locations, or times. These modified transactions are called 'perturbed samples' (Garreau & von Luxburg, 2020). LIME would then ask the complex AI model to make predictions on these perturbed samples. The AI model would classify each perturbed sample as either fraudulent or legitimate. Next, LIME would take the perturbed samples and their corresponding predictions from the complex AI model and try to fit a simpler, more interpretable model (e.g., linear regression or decision tree) to them.

The idea is that this simpler model should closely mimic the complex AI model's behavior for transactions similar to the original one. The coefficients or decision rules of this simpler model are then used to explain the complex AI model's prediction on the original transaction. For example, if the simpler model heavily weighs the transaction amount and location in determining fraud risk, LIME would highlight these features as important factors in

the AI model's original prediction. To sum it up, both SHAP and LIME can explain individual predictions rather than the model's overall behavior. This can be particularly useful in fraud detection, where investigators need to understand the specific factors that led to a particular transaction being flagged as suspicious.

Integrating a conversational AI system with a case-based reasoning (CBR) system is also a powerful approach to provide explanations for fraud detection since it uses knowledge from previously solved problems (cases) to solve new, similar problems (Rudin et al., 2022). When a fraud investigator asks the conversational AI assistant to explain its fraud prediction, the AI would query the integrated CBR system. The CBR system would then search its database of historical fraud cases for those most similar to the current suspicious transaction. Similarity can be based on features like transaction amount, location, customer profile, etc. If the CBR system finds a highly similar historical case (e.g., a previous incident where a customer's account was compromised, and the fraudster transferred a large sum to a foreign account), it would retrieve the solution (explanation) associated with this historical case. The retrieved explanation is then adapted to the current case. The conversational AI assistant might say,

"This transaction is similar to a previous fraud case (case ID: xxx) where the customer's account was compromised. In that case, the large transaction amount and foreign destination were key indicators of fraud."

The conversational AI could also provide additional details:

"In the historical case, the fraudster gained access to the customer's online banking credentials through a phishing scam. The money was transferred to a mule account in Country X before being moved elsewhere. Investigators found that the IP address used to initiate the transfer was suspicious. Similar red flags are present in the current transaction."

If the investigator confirms the AI's prediction and the adapted explanation is valid, this new case and its explanation would be added to the CBR system's case base for future reference (Rudin et al., 2022). The investigator can also probe further the additional detals provided by the AI and ask it to compare and contrast the current case with other similar historical cases, gaining insights into common patterns and potential investigative actions.

Counterfactual explanations are also a powerful technique since they could help investigators understand what factors contributed to a transaction being flagged as suspicious, and how modifying these factors would change the conversational AI's prediction (Byrne, 2019). Suppose the conversational AI assistant flags a credit card transaction as potentially fraudulent due to its high amount ($5,000) and the fact that it originated from a foreign country (Country X) that the cardholder has never visited before. When the fraud investigator asks the AI to explain its decision, the AI assistant would provide a counterfactual explanation:

"The transaction was flagged as suspicious primarily due to the high amount and the unusual transaction location. If the transaction amount had been $500 instead of $5,000, and if it had originated from the cardholder's home country instead of Country X, the transaction would have been classified as legitimate with 90% confidence."

This counterfactual explanation highlights the two key features (amount and location) that led to the fraud prediction, and shows how changing these features would have resulted in a different prediction.

The investigator could then ask follow-up questions to explore other scenarios:

"What if the transaction amount was $5,000 but it originated from the cardholder's home country?"

The conversational AI assistant might respond something along the lines of:

"In that case, the transaction would have been flagged as suspicious with 60% confidence, due to the high amount alone. The unusual location further increased the suspicion score."

Thus, counterfactual explanations provide a clear way to understand the key factors influencing a model's prediction and allow investigators to explore 'what-if' scenarios and gain insights into the model's decision boundaries (Guidotti, 2022).

Presenting the conversational AI's decision-making process in a visual, intuitive format, investigators can quickly grasp the main reasons behind each prediction and explore the underlying data (Ahn & Lin, 2019). For example, if the conversational AI assistant flags a series of credit card transactions as potentially fraudulent, the fraud investigator can ask the AI to generate a dashboard with the following visuals to understand the key factors contributing to these predictions:

i. *Feature Importance Chart:* A bar chart (see Figure 1) displaying the relative importance of each input feature (e.g., transaction amount, location, time, merchant category) in the AI's fraud prediction can aid the investigator in seeing at a glance that certain features (e.g., transaction amount and location) were the top contributors.

< Insert Figure 1 here >

ii. *Decision Tree Diagram:* A simplified decision tree diagram (see Figure 2) showing the main decision paths leading to the fraud classification with each node in the tree representing a key decision factor, and with branches representing different feature values. The investigator can trace the path of a specific transaction and understand the combination of factors that led to its classification.

< Insert Figure 2 here >

iii. *Geospatial Map:* A map visualization (see Figure 3) plotting the location of each flagged transaction, color-coded by fraud risk score can help the investigator identify geographic patterns or hotspots that the AI has learned to associate with fraudulent activity.

< Insert Figure 3 here >

iv. *Parallel Coordinates Plot:* This plot (see Figure 4) allows the investigator to explore relationships between multiple transaction features. Each line represents a transaction, with its feature values plotted along parallel axes. The investigator can highlight transactions classified as fraudulent and look for common patterns or outliers.

< Insert Figure 4 here >

v. *Feature Distribution Chart* (Histogram or Density Plot): For each key feature, a histogram (see Figure 5) or density plot showing the distribution of values for legitimate vs. fraudulent transactions can assist the investigator to see how the conversational AI assistant has learned to distinguish between the two classes based on these feature distributions.

< Insert Figure 5 here >

Fraud investigators can also compare the AI's predictions with human judgments and provide feedback to refine the model.

Further, fraud investigators can use disparate impact analysis to assesses whether the AI model's predictions disproportionately affect certain protected groups (e.g., based on race, gender, age) compared to others. Assume a bank using an AI-powered fraud detection model to flag potentially fraudulent credit card transactions with the model taking into account various features such as transaction amount, location, time, and the cardholder's demographic information (e.g., age, gender, income level). The fraud investigator can ask the conversational AI to gather a representative dataset of credit card transactions, including the relevant features and the actual fraud labels, while ensuring that the dataset includes demographic information for each cardholder. The investigator can then apply the fraud detection model to the dataset and obtain the predicted fraud probabilities for each transaction. Then, the investigator can divide the dataset into protected groups based on sensitive attributes (e.g., gender, age

brackets), and calculate the proportion of transactions flagged as potentially fraudulent by the model for each group.

Lastly, the investigator can compare the proportion of flagged transactions between the protected groups. If the proportion for one group is significantly lower than the other (e.g., below a certain threshold), it may indicate disparate impact. For example, if the model flags 20% of transactions by male cardholders as potentially fraudulent, but only 5% of transactions by female cardholders, it suggests that the model might be disproportionately impacting male cardholders.

Equality of opportunity focuses on ensuring that the model's performance is similar across different protected groups. It can be assessed using metrics like false positive rates or FPR (the proportion of non-fraudulent transactions incorrectly flagged as fraudulent by the model) and false negative rates or FNR (the proportion of actual fraudulent transactions that the model fails to flag) for each group. Using the same dataset as before, the investigator can split it into protected groups based on sensitive attributes and ask the conversational AI to compare the FPR and FNR across the protected groups. If there are significant disparities, it indicates that the model's performance is not equitable across those groups. For instance, if the model has a higher FPR for transactions by younger cardholders compared to older ones, it means that younger cardholders are more likely to be incorrectly flagged for fraud, potentially leading to unfair scrutiny or inconvenience.

4. Conclusion

The integration of conversational AI, such as ChatGPT or Claude, into fraud detection, prevention, and investigation processes represents a paradigm shift in the battle against financial crimes. By leveraging the power of natural language processing and cognitive

computing, financial institutions and regulatory bodies can enhance their AML compliance efforts, streamline investigations, and stay ahead of evolving fraud tactics.

Key Insights:

1. *Enhanced Efficiency:* Conversational AI systems can significantly reduce the time and resources required for tasks such as transaction monitoring, alert triage (reviewing, analyzing, and grouping alerts based on their potential impact and urgency), and regulatory reporting. As illustrated in my scenarios, these systems can automate routine tasks, allowing human investigators to focus on more complex, high-risk cases.
2. *Improved Accuracy:* By leveraging vast amounts of data and advanced pattern recognition capabilities, AI systems can identify subtle indicators of fraud that might be missed by traditional rule-based systems or human analysts. This can lead to higher detection rates and fewer false positives.
3. *Adaptive Learning:* Unlike static rule-based systems, AI models can continuously learn and adapt to new fraud patterns, helping financial institutions stay ahead of evolving criminal tactics.
4. *Regulatory Compliance:* Conversational AI can assist in ensuring consistent adherence to regulatory requirements, from KYC processes to the timely filing of Suspicious Activity Reports (SARs).
5. *Investigative Support:* These systems serve as powerful assistants to human investigators, providing contextual information, historical comparisons, and suggesting lines of inquiry that can accelerate and enhance the investigation process.

However, the successful implementation of conversational AI requires a holistic approach, addressing challenges related to data quality, bias, explainability, and continuous learning and adaptation. Some challenges and considerations include:

1. *Data Quality and Privacy:* The effectiveness of AI systems is heavily dependent on the quality and completeness of training data. Ensuring data accuracy while adhering to privacy regulations remains a critical challenge.
2. *Explainability and Transparency:* As AI systems become more complex, ensuring their decision-making processes are transparent and explainable to regulators, auditors, and customers becomes increasingly important.
3. *Bias Mitigation:* There is an ongoing need to monitor and mitigate potential biases in AI systems to ensure fair and equitable treatment of all customers and transactions.
4. *Regulatory Acceptance:* As AI systems take on more critical roles in fraud detection and AML processes, gaining regulatory acceptance and demonstrating compliance will be crucial.
5. *Human-AI Collaboration:* Striking the right balance between AI automation and human oversight remains a key consideration in maintaining effective and responsible fraud management practices.

Looking ahead, several trends are likely to shape the future of AI in fraud management:

1. *Advanced NLP Capabilities:* As natural language processing technologies continue to advance, we can expect even more sophisticated interactions between AI systems and human investigators, potentially leading to more context-aware fraud detection.
2. *Integration of Multiple AI Technologies:* The combination of conversational AI with other technologies such as computer vision, blockchain, and quantum computing may open new frontiers in fraud detection and prevention.
3. *Cross-Institution Collaboration:* AI systems could facilitate more effective information sharing between financial institutions, enhancing the collective ability to detect and prevent financial crimes.

4. *Regulatory Technology (RegTech):* The development of AI-powered regulatory compliance tools is likely to accelerate, potentially transforming how financial institutions interact with regulators.
5. *Ethical AI Frameworks:* As AI systems become more prevalent in critical decision-making processes, the development and adoption of ethical AI frameworks will be crucial to ensure responsible and fair use of these technologies.

As with any transformative technology, careful consideration and responsible implementation are important to realizing the full potential of conversational AI in fraud management. Baran Ozkan, Co-Founder and CEO of Flagright, quotes: "*The future of AML lies in the seamless integration of AI and human expertise. We're already witnessing a paradigm shift in how investigations are conducted. As regulatory landscapes evolve, so will the role of conversational AI in compliance and fraud prevention. We must bring the best of both worlds from machine and human intelligence to drive meaningful, frictionless change.*" (Ozkan, 2024).

References:

Ahn, Y., & Lin, Y. R. (2019). Fairsight: Visual analytics for fairness in decision making. *IEEE transactions on visualization and computer graphics*, *26*(1), 1086-1095. DOI: 10.1109/TVCG.2019.2934262

Alabi, R. O., Elmusrati, M., Leivo, I., Almangush, A., & Mäkitie, A. A. (2023). Machine learning explainability in nasopharyngeal cancer survival using LIME and SHAP. *Scientific Reports, 13*(1), 8984. DOI: 10.1038/s41598-023-35795-0

Ariza-Garzón, M. J., Arroyo, J., Caparrini, A., & Segovia-Vargas, M. J. (2020). Explainability of a machine learning granting scoring model in peer-to-peer lending. *IEEE Access, 8,* 64873-64890. DOI: 10.1109/ACCESS.2020.2984412

Association of Certified Fraud Examiners. (2023). *Occupational Fraud 2022: Report to the Nations.* Retrieved from https://legacy.acfe.com/report-to-the-nations/2022/

Barocas, S., & Selbst, A. D. (2016). Big data's disparate impact. *California. Law Review*, *104*, 671. DOI: 10.2139/ssrn.2477899

Bell, A., Fonseca, J., Abrate, C., Bonchi, F., & Stoyanovich, J. (2024). Fairness in Algorithmic Recourse Through the Lens of Substantive Equality of Opportunity. *arXiv preprint arXiv:2401.16088*. DOI: 10.48550/arXiv.2401.16088

Byrne, R. M. (2019). Counterfactuals in Explainable Artificial Intelligence (XAI): Evidence from Human Reasoning. In Proceedings of the *Twenty-Eighth International Joint Conference on Artificial Intelligence* (IJCAI), 6276-6282.

Das, A., & Rad, P. (2020). Opportunities and challenges in explainable artificial intelligence (xai): A survey. arXiv preprint arXiv:2006.11371. DOI: 10.48550/arXiv.2006.11371

Dwivedi, R., Dave, D., Naik, H., Singhal, S., Omer, R., Patel, P., ... & Ranjan, R. (2023). Explainable AI (XAI): Core ideas, techniques, and solutions. ACM Computing Surveys, 55(9), 1-33. DOI: 10.1145/3561048

Flagright (2024). Revolutionizing Financial Crime Investigations with AI Forensics: A Flagright Case Study.

Garreau, D., & von Luxburg, U. (2020). Explaining the explainer: A first theoretical analysis of LIME. In Proceedings of the *Twenty Third International Conference on Artificial Intelligence and Statistics, 108,* 1287-1296. Available at https://proceedings.mlr.press/v108/garreau20a.html

Guidotti, R. (2022). Counterfactual explanations and how to find them: literature review and benchmarking. *Data Mining and Knowledge Discovery,* 1-55. DOI: 10.1007/s10618-022-00831-6

H3M Analytics, Inc. (2023). Unlocking the Power of ChatGPT for AML Compliance. Retrieved from H3M.IO Blog.

Hillig, S., & Müller, R. (2021). How do conversational case-based reasoning systems interact with their users: a literature review. *Behaviour & Information Technology*, *40*(14), 1544-1563. DOI: 10.1080/0144929X.2020.1767207

Kurt, T. (2024, April 3). *ChatGPT and LLM models in general will significantly enhance the learning curve for new analysts learning to fight financial crime, and minimize their time spend for the related complex investigations.* [Email].

MacCarthy, M. (2018). Standards of Fairness for Disparate Impact Assessment of Big Data Algorithms. *Cumberland Law Review, 48,* 67. DOI: 10.2139/ssrn.3154788

Molnar, C. (2022). *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable (2nd ed.).* Available at https://christophm.github.io/interpretable-ml-book/

Neves, I., Folgado, D., Santos, S., Barandas, M., Campagner, A., Ronzio, L., Cabitza, F., & Gamboa, H. (2021). Interpretable heartbeat classification using local model-agnostic explanations on ECGs. *Computers in Biology and Medicine, 133,* 104393. DOI: 10.1016/j.compbiomed.2021.104393

Ozkan, B. (2024, April 10). *The future of AML lies in the seamless integration of AI and human expertise. We're already witnessing a paradigm shift in how investigations are conducted. As regulatory landscapes evolve, so will the role of conversational AI in compliance and fraud prevention. We must bring the best of both worlds from machine and human intelligence to drive meaningful, frictionless change.* [Email].

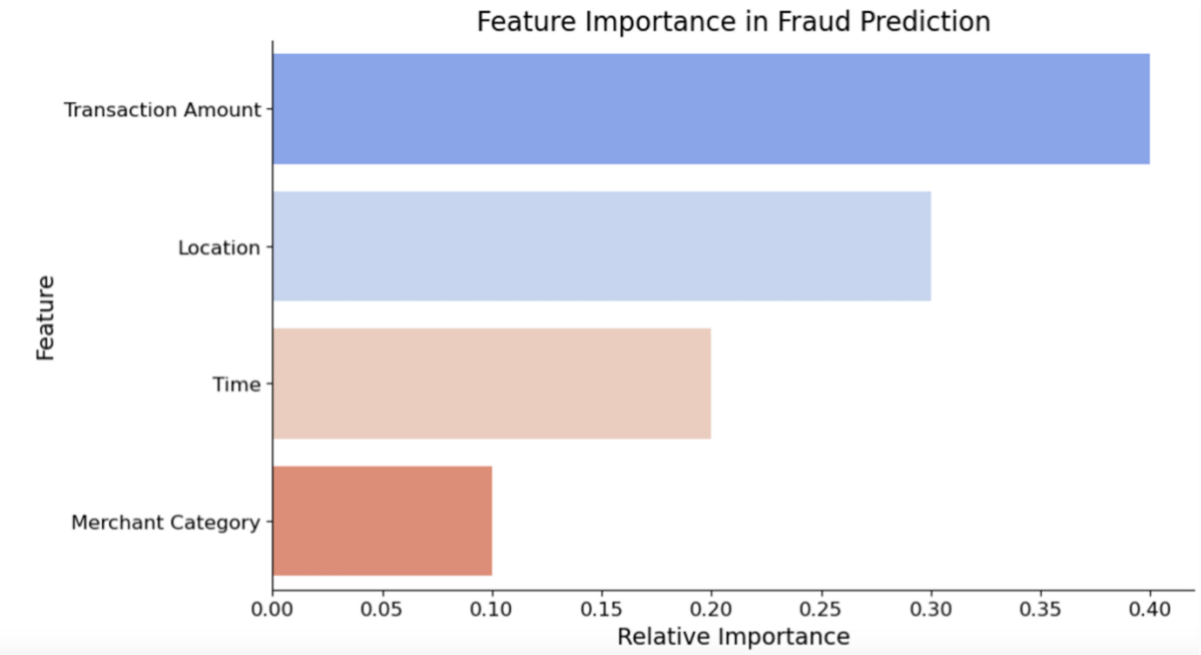
Parikh, R. B., Teeple, S., & Navathe, A. S. (2019). Addressing bias in artificial intelligence in health care. JAMA, 322(24), 2377-2378. DOI: 10.1001/jama.2019.18058

Rudin, C., Chen, C., Chen, Z., Huang, H., Semenova, L., & Zhong, C. (2022). Interpretable machine learning: Fundamental principles and 10 grand challenges. *Statistic Surveys, 16,* 1-85. DOI: 10.1214/21-SS133

Shukla, S., George, J. P., Tiwari, K., & Kureethara, J. V. (2022). Data security. In Data ethics and challenges (pp. 41-59). Springer Singapore. DOI: 10.1007/978-981-19-0752-4

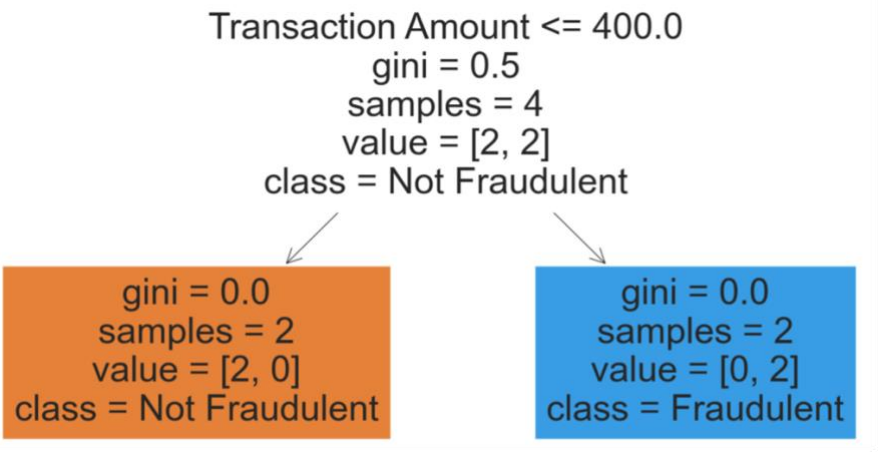
Tachepun, C., & Thammaboosadee, S. (2020, July). A data masking guideline for optimizing insights and privacy under GDPR compliance. In *Proceedings of the 11th International Conference on Advances in Information Technology* (pp. 1-9). Association for Computing Machinery. DOI: 10.1145/3406601.3406627

Figure 1: Feature Imporance Chart



Note:
The feature importance chart provides insights into which features the fraud detection model considers most influential in making its predictions. The chart shows the relative importance of four features: Transaction Amount, Location, Time, and Merchant Category. Importance is measured from 0 to 1, with higher values indicating greater importance. Transaction Amount has the highest importance at around 0.40 with Location coming at a close second at around 0.30. This suggests the amount of a transaction and location amounts are the strongest indicators of whether a transaction might be fraudulent, while time and merchant category are less important (though they provide supporting evidence). In summary, if this model flags a transaction as likely fraudulent, we can infer that a highly unusual transaction amount was probably the biggest factor, followed by a suspicious location.

Figure 2: Decision Tree Diagram



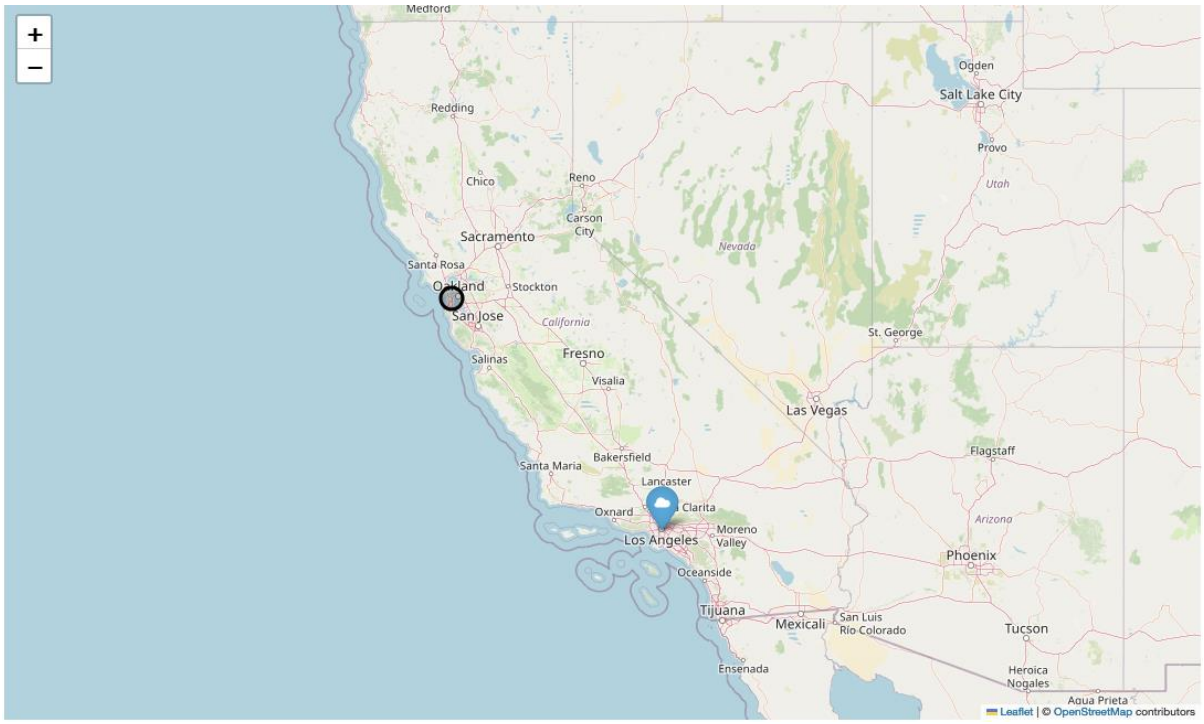
Note:
This simple decision tree diagram uses the transaction amount as the key feature to predict fraud. It has learned from the training data that transactions above 400.0 are likely to be fraudulent, while those below or equal to 400.0 are likely to be legitimate. The decision tree is thus trying to classify transactions as either 'Fraudulent' or 'Not Fraudulent' based on the transaction amount.

It does this by splitting the data at a certain threshold value of the transaction amount. The first node (at the top) represents all the transaction data before any splitting. It shows (a) the splitting criterion: Transaction Amount <= 400.0; and (b) the Gini impurity of the node: 0.5. There are 4 total samples (transactions) in this node. The average transaction amount is [2, 2], meaning there are 2 transactions of each class (Fraudulent and Not Fraudulent).

A Gini impurity score give an idea of how good a split is by how mixed the classes are in the two groups created by the split, and a lower Gini score is better because it indicates a lower impurity. A Gini impurity of 0 means the node is completely pure, i.e., all samples in that node belong to the same class (either all 'Fraudulent' or all 'Not Fraudulent'). A Gini impurity of 1 would mean the node is completely impure, i.e., the samples are evenly distributed across classes. In a binary classification problem like this, it would mean the node has an equal number of 'Fraudulent' and 'Not Fraudulent' samples. This is the worst case, as the tree cannot make a confident classification for samples in this node. The root node has a Gini impurity of 0.5, which means it's equally likely for a randomly selected sample to be 'Fraudulent' or 'Not Fraudulent'. This makes sense, as there are 2 samples of each class in this node.

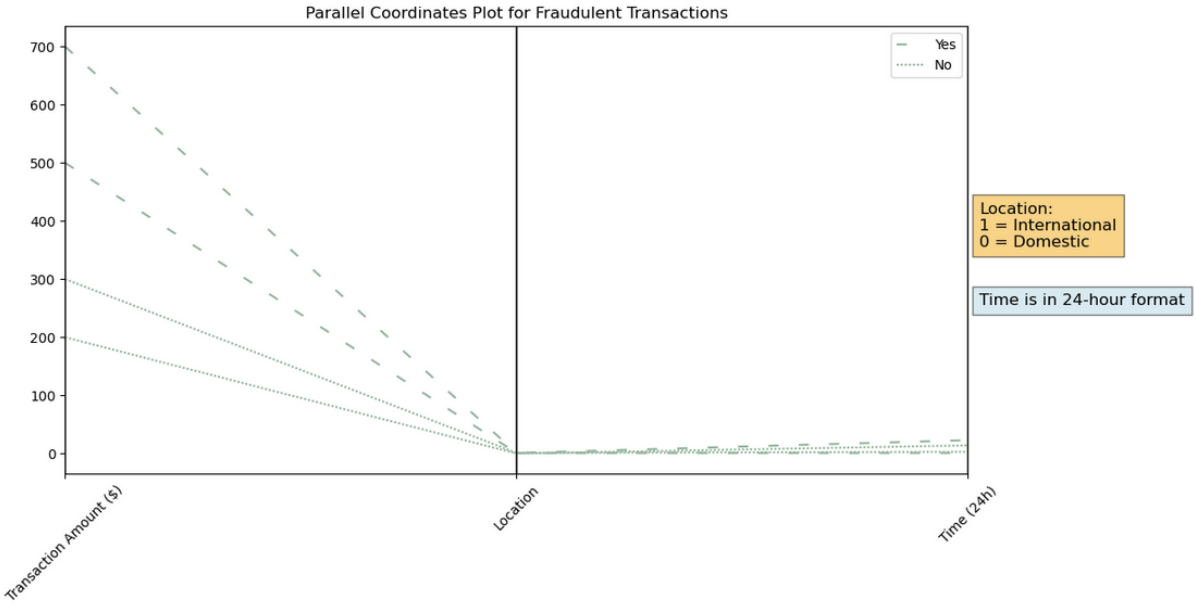
If a transaction's amount is less than or equal to (greater than) 400.0, it goes to the left (right) child node. Each of these nodes have Gini impurities of 0.0, which suggests they are pure nodes (all samples belong to one class). There are 2 samples (transactions) in each of these nodes. The average value in the left node is [2, 0], meaning both transactions are 'Not Fraudulent'. So, if a transaction's amount is less than or equal to 400.0, the tree classifies it as "Not Fraudulent". The average value in the right node is [0, 2], meaning both transactions are 'Fraudulent'. So, if a transaction's amount is greater than 400.0, the tree classifies it as "Fraudulent".

Figure 3: Geospatial Map



Note:
This geospatial map provides an example visualizing the geographic distribution of potential fraud risks. Each marker on the map represents a specific location, and its shape indicates the associated fraud risk score. Circle markers signify locations with a high fraud risk (risk score > 0.5), while the Cloud markers with an icon represent locations with a low fraud risk (risk score ≤ 0.5). In this example, there are two markers: one in the San Francisco Bay Area (circle) and one in Los Angeles (cloud). This suggests that the location in the Bay Area has a higher risk of fraudulent activity compared to the location in Los Angeles. This visual representation allows fraud investigators to focus their efforts on high-risk areas and allocate resources accordingly.

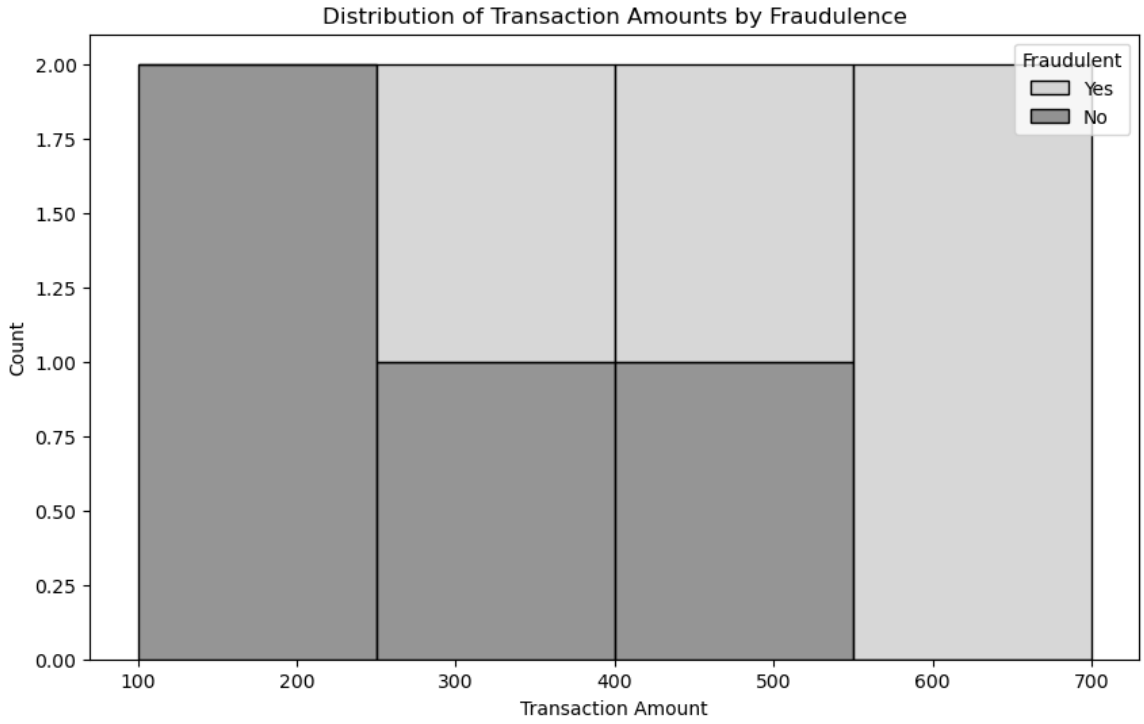
Figure 4: Parallel Coordinates Plot



Note:
This parallel coordinates plot allows investigators to visually identify patterns and relationships between the features and the fraudulent classification. The plot consists of three parallel axes, each representing a different feature: Transaction Amount, Location and Time. Each line in the plot represents a single transaction. The line intersects each axis at the corresponding feature value for that transaction. The pattern of the lines indicates whether the transaction is classified as fraudulent or not (dashed lines = Yes; dotted lines = No). The Transaction Amount axis shows the monetary value of each transaction. The Location axis is binary, with 1 representing international transactions and 0 representing domestic transactions. The Time axis represents the time of the transaction in military format (0-23 hours).

We can observe that the two fraudulent transactions (dashed lines) have higher transaction amounts compared to the non-fraudulent transactions (dotted lines). Both fraudulent transactions are international (intersecting the Location axis at 1), while one of the non-fraudulent transactions is domestic (intersecting at 0). The fraudulent transactions occur at different times (23 and 1), suggesting that time alone may not be a strong indicator of fraud in this limited dataset. Thus, in this example, higher transaction amounts and international locations seem to be more associated with fraudulent activity.

Figure 5: Feature Distribution Chart



Note:

This feature distribution chart is a stacked histogram that visualizes the distribution of transaction amounts by fraudulence status. Each bar represents a range of transaction amounts on the x-axis, with the count of transactions on the y-axis. From the chart, we can see two distinct shades representing different classes: one for 'No' (not fraudulent) and another for 'Yes' (fraudulent). The bins (horizontal intervals on the x-axis) aggregate transaction amounts into groups, making it easier to observe patterns. The stacked nature of the histogram allows us to compare the relative proportions of fraudulent and non-fraudulent transactions within each transaction amount bin. We can observe if certain amount ranges have a higher or lower proportion of fraudulent transactions. From this chart, we can see that fraudulent transactions seem to be more prevalent in the higher transaction amount ranges. Non-fraudulent transactions appear to be more evenly distributed across the transaction amount ranges. Thus, this chart suggests that transactions with higher amounts might be more likely to be fraudulent compared to transactions with lower amounts. However, it's important to note that this is just one variable, and fraud detection typically involves analyzing multiple features.